Instituto Superior Técnico

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Data Analytics for Smart Grids

Power System State Estimation

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The Base Problem

Problem Formulation

Power system state estimation is a fundamental tool for modern energy management systems, providing a reliable estimate of the system operating state despite noisy and incomplete measurements. Consider a 5-bus radial distribution network with slack bus at node 5. The objective is to estimate the state vector $\mathbf{x} \in \mathbb{C}^n$ containing complex voltages at all non-slack buses:

$$\mathbf{x} = \begin{bmatrix} V_1 & V_2 & V_3 & V_4 \end{bmatrix}^{\mathrm{T}} \tag{1}$$

Mathematical Formulation

To find this state vector, the wls estimator is used to minimize the sum of the squared measurement residuals, scaled by their respective accuracies. The estimated state vector $\mathbf{x}$ is obtained by solving the equations:

$$\mathbf{x} = (H^T W H)^{-1} H^T W \mathbf{z} \tag{2}$$

Performance Analysis

Scenario	MSE	max $\lvert\varepsilon\rvert$ (%)	Key Characteristic
1º	5.92×10^{-2}	27.3	Underdetermined, no weighting
2º a)	1.03×10^{-9}	0.007	Full redundancy, no noise
2º a)	2.98×10^{-5}	0.77	Full redundancy, uniform weights
2º b)	2.02×10^{-6}	0.19	Optimal weighting

Table 1: State Estimation

Table 1 shows why having extra data and correct weighting is so important. In Scenario 1, the addition of multiplicative noise with no redundancy and a $\sigma = 0.5$ lead to a high and unacceptable error of 27.3%, miles above any other estimation performed.

While adding pseudo-measurements in Scenario 2a makes the system work (really well in a case with zero noise), the results when noise is added only get really good in Scenario 2b. By using the weight matrix (W), the system finally accounts for the fact that professional meters are much more reliable than rough estimates, giving weights to each parameter based on error. This adjustment drops the error

significantly, proving that the wls method doesn't just use the extra data, it actually filters out the noise to give a much cleaner result.

Additionally, a set of Monte Carlo simulations was run to view how the mismatch between the real values of I_{12} and I_{54} and the estimated values evolved with noise level in the pseudo-parameter part of the data. This was done for two cases, the first one using a weight matrix built with the squared inverse of the error values for each line.

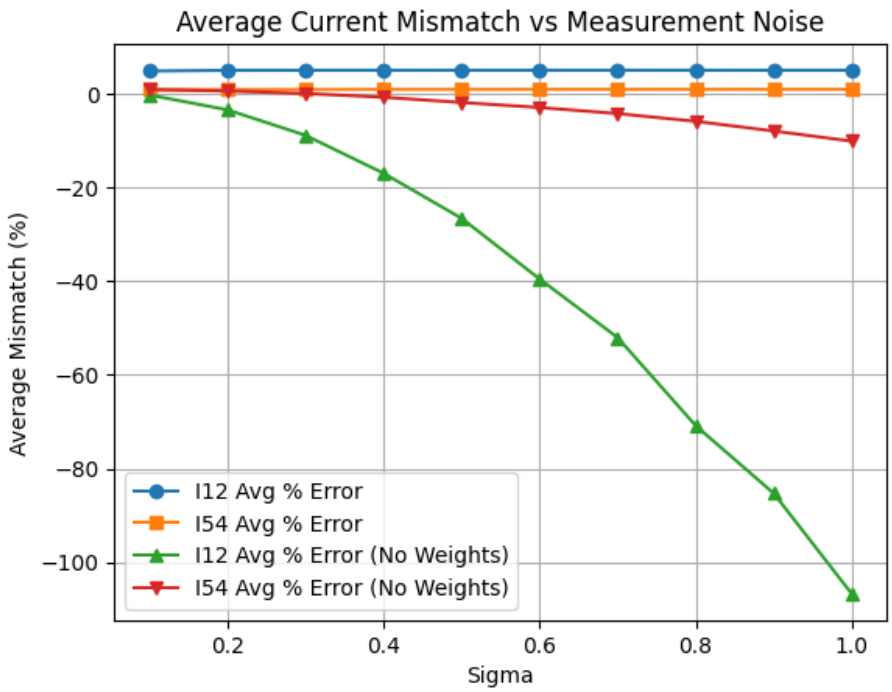

Figure 1: Average Current Mismatch vs. Measurement Noise

As expected, given that the weight matrix used contains a weighting factor based on error, increasing error doesn't affect mismatch, especially since the two variables for which the mismatch is being calculated were determined before the addition of pseudo-measurements. With the removal of the weight matrix, mismatch shoots up as error increases, especially in I_{12}. This can indicate that the estimation for I_{54} is more stable under high levels of noise.

Challenges

Optimal Pseudo-Measurement Selection

In practical power system state estimation, the number of available real time measurements is often insufficient to ensure system observability, necessitating the inclusion of pseudo-measurements derived from historical load data or generation forecasts. However, not all pseudo-measurements contribute equally to estimation accuracy. Their effectiveness depends on network topology, measurement place-

ment and the resulting numerical conditioning of the measurement Jacobian matrix. This challenge addresses the critical question of measurement selection: given the constraint of choosing only two pseudo-measurements from four available power injection measurements, which combination maximizes estimation accuracy?

To answer this question, all possible combinations of pseudo-measurements were tested in a Monte-Carlo simulation environment, where 1000 runs were performed for each combination. In these calculations, the H_x matrix was constructed with the first four rows intact, since they correspond to the real measurements, and the remaining two rows set as every possible combination of two of the remaining four pseudo-parameters, totaling six possibilities. Then, the average traces of the covariance matrices were computed, and all six combinations were compared:

Combination	Average Trace
S_{12}	1.005×10^{-6}
S_{13}	1.262×10^{-6}
S_{14}	3.528×10^{-6}
S_{23}	1.017×10^{-5}
S_{24}	3.799×10^{-6}
S_{34}	4.178×10^{-6}

As we can see, combination S_{12} yielded the smallest trace, with S_{23} trailing close behind. The one thing left to check is wether these combinations lead to a fully determined system. This can be done by checking the rank of the Hx matrix. Since we're trying to estimate a set of five voltages, a rank of 5 is needed. This determination of rank was done in python, and was positive for S_{12}, making it the best combination of pseudo-measurements.

Topology Error Detection and Meter Malfunction Identification

Another challenge in power system monitoring is telling the difference between real network changes, like a line going down and sensor errors that create bad data. If we see a current reading of zero, we need to know: is the line actually out of service, or is the meter just broken? Getting this right is vital for the grid, as a mistake could lead to the wrong control actions or missed repairs. In this task, we use a hypothesis testing approach based on wls estimation with different weight matrices to solve this.

Figure 2: ainda a fazer

To test how the system handles anomalies, we looked at a specific case where a meter unexpectedly reports $0A$ for the current I_{12}. We set up two competing ideas: Hypothesis A assumes the line 1-2 is actually down ($y_{12} = 0$), meaning the zero reading is correct. Hypothesis B assumes the line is still fine ($y_{12} \neq 0$), but the meter itself is failing and giving a false zero. To decide which is true, the estimator calculates the weighted sum of squared residuals:

$$J(x) = [\mathbf{z} - h(\mathbf{x})]^T W [\mathbf{z} - h(\mathbf{x})] \tag{3}$$

Initially, we tested these using an identity matrix, treating every measurement as equally reliable. Under this setup, the error for Hypothesis A was 1.7558×10^{-5}, while Hypothesis B was 1.2852×10^{-2}. Although Hypothesis B has a higher error, the gap isn't wide enough to confidently pick a winner. Without proper weights, the algorithm trusts the faulty $0A$ reading as much as the precise voltage data, forcing the calculated voltages at buses 1 and 2 to "bend" just to justify that zero current. This effectively spreads the error across the network instead of pointing to the source.

The real observation happens when the power appears once we apply the proper weight matrix. This shift prioritizes high-precision meters over less certain data. As shown in the plots, the error for Hypothesis A becomes 2.8861×10^{-2}, while the error for Hypothesis B explodes to 5.0213×10^{-1}—nearly 17.4 times larger. Because the weight matrix now heavily penalizes any deviation from the accurate voltage readings, the estimator refuses to "fake" the node potentials. Since Ohm's Law says you can't have zero current between two points with different confirmed voltages, Hypothesis B creates a mathematical mistake.

This numerical evidence makes the choice clear: Hypothesis A is the best for this case. The meter

isn't broken; the line 1-2 is physically out of service. This proves that the W_z matrix isn't just a minor tweak for accuracy, it is a mandatory tool for identifying the actual state of the grid and catching bad data.

Phasor Measurement Unit Problem

A phasor measurement unit is a high-speed sensor that provides synchronized, real-time measurements of voltage and current phasors. By using GPS time-stamping, PMUs offer a level of precision and situational awareness that standard meters can't match. In this test, we installed a single PMU on Bus 3, the blind spot of our network, which previously relied on noisy load estimates.

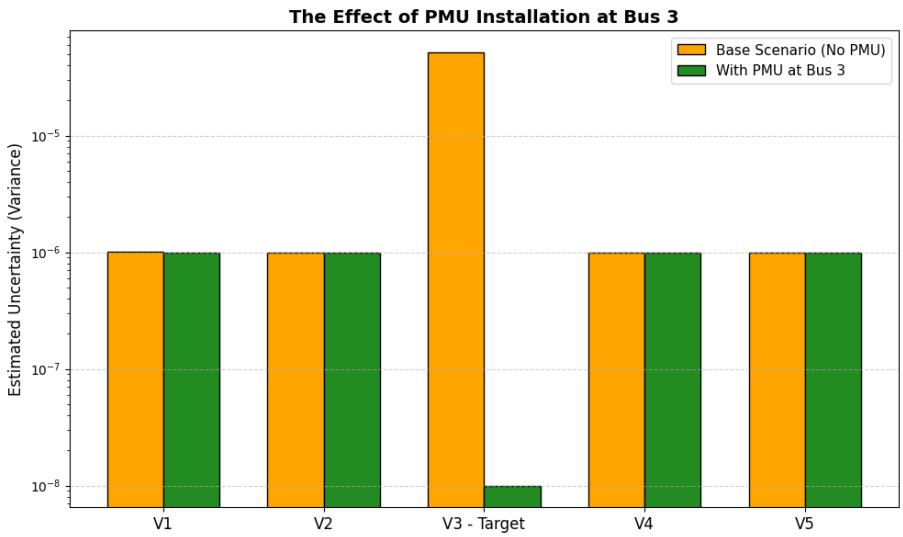

Figure 3: ainda a fazer

As shown in the bar chart, the PMU drastically cuts the uncertainty (V_3) at that specific node. Interestingly, this improvement doesn't spread to the neighboring buses (V_1, V_2, V_4, V_5). This highlights a core principle of the wls estimator: since those buses were already anchored by high-precision sensors, their accuracy was already at its limit. The PMU acts locally, fixing the one weak link without needing to adjust the rest of the grid.

To put a number on this, we looked at the trace of the covariance matrix, which measures the total uncertainty of the system.

- Base Scenario: With only standard meters and noisy estimates for Bus 3, the global uncertainty was 5.5731×10^{-5}.
- With PMU: After adding the PMU (with a high-precision weight reflecting its $\sigma = 0.0001$ accuracy), the uncertainty plummeted to 3.9965×10^{-6}.

This represents a massive 92.8% reduction in total system variance just by upgrading a single node. The chart confirms that this gain is almost entirely localized at Bus 3. Because the rest of the network

was already secured by the weight matrix, the PMU simply neutralized the system's highest point of low observability without disturbing the already solid data elsewhere. This proves that strategic PMU placement, combined with wls, is incredibly efficient at cleaning up grid observability.

Conclusion

This laboratory demonstrated the robustness of the wls state estimator. It was found that data redundancy only adds real value when coupled with a properly calibrated weighting matrix, which is essential to confine the high uncertainty of pseudo-measurements to the system's blind nodes. We also found that the estimator does more than just filter noise; it acts as a tool to isolate bad data and detect physical faults in the system. Finally, the covariance matrix proved to be a good analytical way to decide where to place PMUs.

Bibliography

[1] Pinson, P. [ou Matos, P.] (2026). Losses Prediction (Chap. 2.2). Apontamentos da disciplina de Analítica de Dados para Redes Inteligentes. [Instituto Superior Técnico].